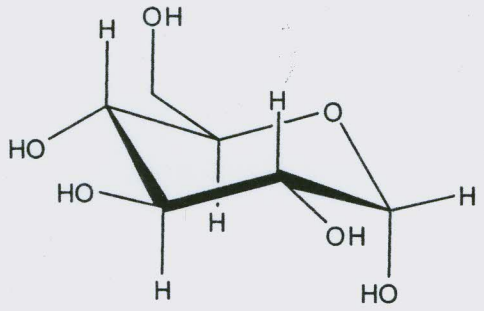
Biochemistry Cumulative Examination

1. Consider D-glucose, shown below in pyranose form. Draw the Fischer projection. (10 points)



2. Draw the disaccharide of glucose (maltose) with the linkage that is found in starch. (15 points)

3. Draw the disaccharide of glucose (cellobiose) with the linkage that is found in cellulose. (15 points)

4. Draw glucose-6-phosphate. (5 points)

5. Show a mechanism (*i.e.* push some arrows) for the conversion of glucose-6-phosphate to fructose-6-phosphate, one of the first steps of glycolysis. In this and the subsequent mechanisms, you may simply show any plausible side chain functionality needed to carry out your proposed reaction without being too specific. (20 points)

6. Show a mechanism for the conversion of fructose-1,6-bisphosphate into glyceraldehyde 3-phosphate and dihydroxyacetone phosphate. (25 points)

7. One of the consequences of diabetes is high blood sugar levels. Diabetics often have glucose attached to the lysines of their proteins as a Schiff's base. Draw the structure of this side chain-sugar adduct. (10 points)